Lecture 2 Introduction to Linux Notes

1. What is an Operating System? An **Operating system(OS)** is something that provides a computer with all the fundamental software it needs which enables you to use the computers hardware in turn providing you the basic tools that make that computer useful.

2. What is a kernel? A **Kernel** is a software component responsible for managing low-level features o a computer, which include managing system hardware,cpu time, memory allocation, and program to program interaction.

3. Which other parts aside from the kernel identify an OS? Other components that Identify as an OS include: **Command line shells**, **Graphical User-Interfaces**, **Utility and Productivity programs** and **Libraries**.

4. What is Linux? **Linux** is a Unix-like operating System that is popular in academic and business environments.

5. What is a Linux distribution? **Linux Distribution** is a complete linux system package made up of these elements: **A Linux Kernel**, **Core Unix tools**, **Supplementary software**, **Startup scripts**, and **A Installer**.

6. List at least 4 Linux characteristics: Linux is **open source**, **free of charge**, **High scalable and customizable**, and lastly you can install Linux on **any system**.

7. What is Debian? Debian is an all-volunteer organization dedicated to developing free software and promoting the ideals of a free software community.

8. List and define the different types of licensing agreements. There are **3 types** of Licensing agreements; First we have **Open Source agreements** where the source code is with the software and the software may be distributed for free or for a fee. Second we have **Close Source agreements** where the software is unchangeable and isn't distributed along with the source code. Lastly we have **Free Software agreements** which is similar to **Open Source** where the software is free along with the source code.

9. What is Free Software? Define the 4 freedoms. The **4 Freedoms** of Free software is **the ability to use the software for any purpose**, **The ability to examine the source code and modify as one sees fit**, and lastly **The ability to redistribute the original software or your own modified version**.

10. What is virtualization? **Virtualization** is the process of creating virtual versions of something.

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